

MATHS 2

ASSIGNMENT 3

Q1. Find the number of Integers between 4 and 100 that end with 3 or 5.

Numbers ending with 3: 13,23,33,43,53,63,73,83,93 (9 numbers)

Numbers ending with 5: 5,15,25,35,45,55,65,75,85,95 (10 numbers)

Total = $9 + 10 = 19$

Q2. Define Extended Pigeonhole Principle and using it show that any 30 people are selected than at least 5 people must have been born on the same day of the week.

If N objects are placed into k boxes, then at least one box contains $\lceil N/k \rceil$ objects.

Here $N=30$, $k=7$

$\lceil 30/7 \rceil = \lceil 4.28 \rceil = 5$

Hence, at least 5 people are born on the same day.

Q3. A box contains 6 red balls,4 white balls and 5 black balls. A person draw 4 balls from the box at random. Find the probability that among the balls drawn there is at least one ball of each color.

Total balls = 15

Total ways = $15C4 = 1365$

Favourable cases (2,1,1 combinations) = 720

Probability = $720/1365 = 48/91$

Q4. A card is drawn from a well shuffled deck of 52 cards and then second card is drawn. Find the probability that one card is a spade and then second card is club if the first card is not replaced.

Favourable cases = 338

Total cases = 52×51

Probability = $338/2652 = 13/102$

Q5. Nitin has 5 Friends. In how many ways can he invite one or more of them to his party?

Total subsets of 5 friends = $2^5 = 32$

Exclude empty set: $32 - 1 = 31$ ways

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Q6. In how many ways 6 Children can be arranged in a line such that,

i. Two particular children of them are always together.

ii. Two particular children of them are never together.

(i) Always together:

$$5! \times 2! = 120 \times 2 = 240$$

(ii) Never together:

$$\text{Total} - \text{together cases} = 720 - 240 = 480$$

Q7. How many four letter words can be format from the letters G,R,O,U,V,P,S. If no letter is to be used more than Once in any word?

$${}^7P_4 = 7 \times 6 \times 5 \times 4 = 840$$

Q8. How many permutation of letters {a,b,c} are there?

$$3! = 6$$

Q9. Find out probability distribution of the number of heads when three coin are tossed.

Total outcomes = 8

$$P(0 \text{ heads}) = 1/8$$

$$P(1 \text{ head}) = 3/8$$

$$P(2 \text{ heads}) = 3/8$$

$$P(3 \text{ heads}) = 1/8$$

Q10. In a survey of 400 students in a school, 100 were found as drinking apple juice, 150 were drinking orange juice and 75 were drinking both apple and orange juice. Find how many students drink neither apple juice nor orange juice.

$$n(A \cup B) = 100 + 150 - 75 = 175$$

$$\text{Neither} = 400 - 175 = 225 \text{ students}$$